

Math 125

Integration and Applications

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Office hours TBA.

Lectures MWF

Quiz sections TTh

More info on Canvas syllabus

E-textbook: **Calculus: Early Transcendentals**, 9th ed

Purchase by clicking the "**Cengage WebAssign**" link on Canvas.

Required to access WebAssign, which contains homework.

My recommendation is to read each textbook section before the corresponding lecture, according to the schedule found on the Math 125 materials page.

Many homework problems are based on textbook examples, so it is a good place to look if you are stuck.

AI policy:

AI usage on homework is not forbidden, but strongly discouraged. Homework is only 5% of your grade, and relying on AI will leave you unprepared for quizzes and exams, which are the vast majority of your grade.

Learning is a physical process. You cannot learn math by watching, just as you can't learn to play an instrument by listening to music.

True or False: Integration is the opposite of Differentiation.

True or False: Math is hard.

Antiderivatives

(4.9)

Recall that the **derivative** of a function $f(x)$ is a function $f'(x)$ that outputs the instantaneous rate of change of f at a value x .

Example: Let $s(t)$ be the position of a particle at time t .

In your previous calculus class, you discussed how to take derivatives of some common functions.

Let $F(x)$ be a function. Suppose we know the derivative of $F(x)$ is $f(x)$, where

$$f(x) = x^2$$

What is $F(x)$?

The functions $F(x) = \frac{1}{3}x^3$, $F(x) = \frac{1}{3}x^3 + 1$, $F(x) = \frac{1}{3}x^3 + 9.9$ are all **antiderivatives** of $f(x) = x^2$. In fact, any function of the form

$$F(x) = \frac{1}{3}x^3 + C,$$

where C is a real number, is an antiderivative of $f(x) = x^2$. The expression $F(x) = \frac{1}{3}x^3 + C$ is called the **general form** for the antiderivative of $f(x) = x^2$. It is not a single function, but an **infinite family** of functions.

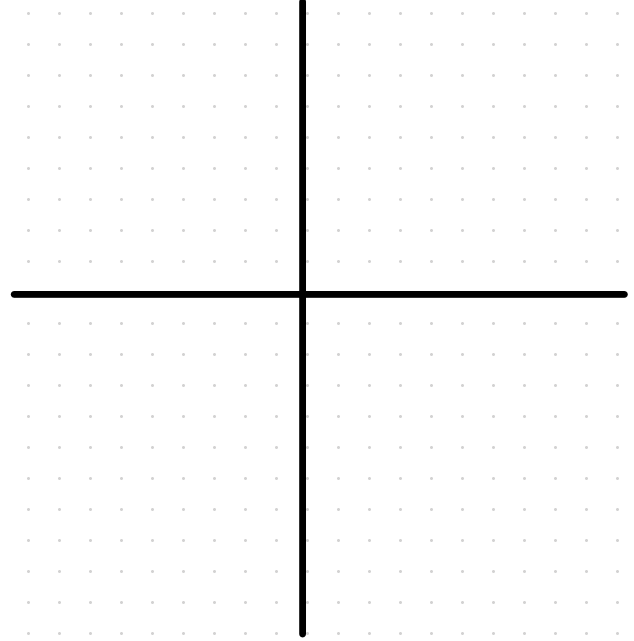
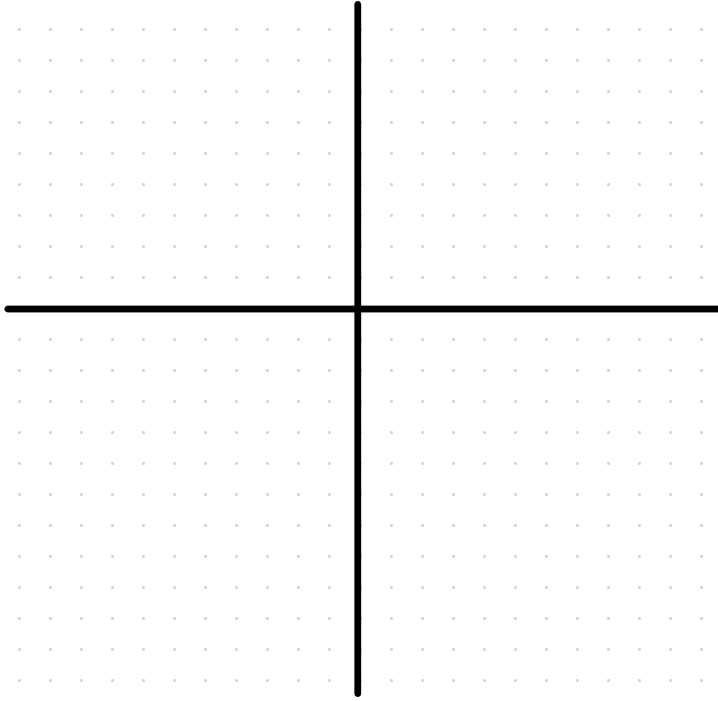
The following theorem tells us that the above "general form" actually covers all antiderivatives.

Theorem: Let $f(x)$ be a function defined on an interval, and let $F(x)$ be **any** antiderivative of f on that interval. Then **all** antiderivatives of f on that interval are of the form.

$$G(x) = F(x) + C$$

where C is a real constant.

Visualizing all the antiderivatives of $f(x)=x^2$.



Find the general form for an antiderivative of

$$f(x) = 5x^3 + \sin x + 4$$

Find the general form for an antiderivative of

$$f(x) = \frac{x^2 + x + 1}{\sqrt{x}}$$

Suppose $f'(x) = 2x^2 + x$ and $f(0) = 1$. What is $f(x)$?

Suppose $g''(x) = x^2$, $g'(0) = 1$, $g(0) = 2$. What is $g(x)$?

Ignoring air resistance, a falling object accelerates at 9.8 m/s^2 . An object is dropped from a height of 10 m . How fast is it moving when it hits the ground?

What is the general form for an antiderivative of

$$f(x) = \frac{1}{x} ?$$

Theorem: Let $f(x)$ be a function defined on an interval, and let $F(x)$ be any antiderivative of f on that interval. Then all antiderivatives of f on that interval are of the form.

$$G(x) = F(x) + C$$

where C is a real constant.

On the interval $(0, \infty)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln x + C$$

On the interval $(-\infty, 0)$, the general form of an antiderivative of $\frac{1}{x}$ is

$$\ln(-x) + D$$

You will often see these combined as

$$\ln |x| + C$$

This should be understood as the form of all antiderivatives on $(0, \infty)$, or all antiderivatives on $(-\infty, 0)$, but not all antiderivatives defined on $(-\infty, 0) \cup (0, \infty)$ are of this form.